

Skills Summary

Canceled-Factor Holes vs Vertical Asymptotes

Use this reference while completing your worksheet or when studying.

Skill 1 — Factor and Reduce a Rational Expression

Always factor before you decide anything. A zero of the original denominator is a break in the domain; *which kind* of break depends on whether that factor cancels.

- Difference of squares: $x^2 - c^2 = (x - c)(x + c)$
- Monic trinomial: find the factor pair of the constant that sums to the middle coefficient
- Reducing loses information — carry the excluded x -values forward yourself

Example: Reduce $\frac{x^2 - 4}{x^2 - x - 6}$ and state the restriction.

Step 1. Both parts are quadratic and the question is which factors are shared, so factor completely before cancelling anything.

Step 2. $\frac{(x - 2)(x + 2)}{(x - 3)(x + 2)} = \frac{x - 2}{x - 3}$, valid for $x \neq -2$ and $x \neq 3$.

Try it: Reduce $\frac{x^2 - 1}{x^2 + 4x - 5}$.

$\frac{x + 1}{1 + x}$:check

Skill 2 — Locate Holes and Vertical Asymptotes

After reducing:

- Factor that **cancelled** $\rightarrow$ *hole* at its zero. Height = the *reduced* form evaluated there.
- Factor that **survived** $\rightarrow$ *vertical asymptote* at its zero.
- Reduce first, substitute once. Mixing the original numerator with the reduced denominator gives a false height of 0.

Example: Height of the hole of $\frac{x^2 - 16}{x^2 - 2x - 8}$ at $x = 4$.

Step 1. The factor $x - 4$ cancels, so the two-sided limit exists and equals the reduced form's value — evaluate the reduced form, never the original.

Step 2. $\frac{(x - 4)(x + 4)}{(x - 4)(x + 2)} = \frac{x + 4}{x + 2}$, so the height is $\frac{4 + 4}{4 + 2} = \frac{4}{3}$.

Try it: Height of the hole of $\frac{x^2 - 1}{x^2 - 3x + 2}$ at $x = 1$.

$\frac{2 - 1}{2 - 3 + 2}$:check

Skill 3 — Tell a Hole from an Asymptote

Count multiplicities, do not just match factors. A shared factor cancels only as many times as it appears on top.

$\frac{(x-c)}{(x-c)^2}$ leaves one $(x-c)$ downstairs $\rightarrow$ *asymptote*. $\frac{(x-c)^2}{(x-c)}$ cancels out fully $\rightarrow$ *hole*.

(!) A perfect-square denominator is the usual trap: $x^2 - 6x + 9 = (x-3)^2$ holds the factor twice.

Example: A student says $\frac{x^2 - 9}{x^2 - 6x + 9}$ has a hole at $x = 3$. Fix it.

Step 1. The factor $x - 3$ appears once on top and twice on the bottom, so it cannot cancel completely — compare multiplicities before deciding.

Step 2. $\frac{(x-3)(x+3)}{(x-3)^2} = \frac{x+3}{x-3}$: one factor survives, so $x = 3$ is a vertical asymptote.

Try it: Reduce $\frac{x^2 - 25}{x^2 + 10x + 25}$ and say what sits at $x = -5$.

check: $\frac{5+x}{5-x}$, asymptote